# Parametric Equations: Integration

Questions 1–22

**Question 1. Prerequisite knowledge: The following questions may involve integrating standard functions or reversing the chain rule.** For each of the following curves defined by the parametric equations, find $\int y\,dx$:

a) $x = 3t^3 - t$ and $y = t^2 + 1$
b) $x = e^t$ and $y = e^{-t} + 1$
c) $x = 3^t$ and $y = 9^t$

**Question 2.** For each of the following curves, where the lower and upper limit ($t_1$ and $t_2$ respectively) of the definite integral have already been found in terms of $t$, find $\int_a^b y\,dx$ giving your answers in exact form:

a) $x = 2t^3 + t$ and $y = t^2 + 4$ between $t_1 = 1$ and $t_2 = 2$
b) $x = t^2$ and $y = e^t$ between $t_1 = 0$ and $t_2 = 2$
c) $x = 2^t$ and $y = 4^t$ between $t_1 = 0$ and $t_2 = 1$
d) $x = -\ln t$ and $y = t^3$ between $t_1 = 0.5$ and $t_2 = 1$

**Question 3.** A curve is defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = \frac{1}{t}$$

for $1 \leq t \leq 2$.

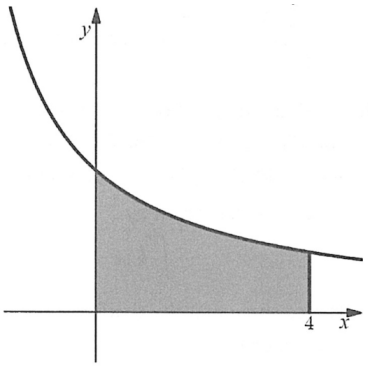
a) Find the coordinates of the curve when $t = 1$ and $t = 2$.

b) Sketch the curve.

c) Find the area enclosed by the curve, the coordinate axes, the lines $x = 1$ and $x = 4$.

**Question 4.** The diagram shows part of a curve defined by

$$x = 2t - 3 \quad \text{and} \quad y = \frac{4}{t}$$




The shaded region is bounded by the curve, the coordinate axes and the line $x = 4$.

a) Find the value of $t$ when $x = 0$ and $x = 4$.

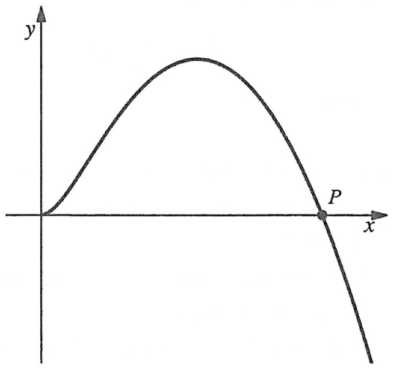
Find the area of the shaded region

b) by using parametric integration.

c) by first finding a cartesian equation for the curve.

**Question 5.** A curve is defined by parametric equations

$$x = t^2 + 2t \quad \text{and} \quad y = 2t^2 - t^3 \quad \text{for } t \geq 0$$




The point $P$ is where the curve meets the $x$-axis at $x = k$. The area $A$ is enclosed by the curve and the $x$-axis is given by

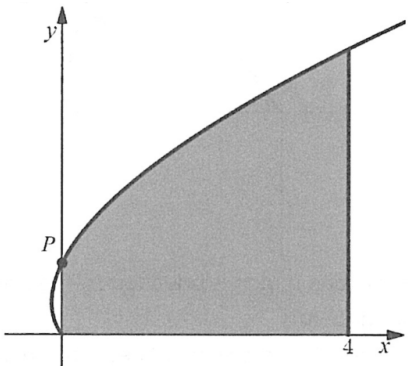
$$A = \int_0^k y \, dx$$

a) Find the value of $k$.
b) Find the area $A$.

**Question 6.** The diagram shows the curve given by

$$x = t^2(t - 1) \quad \text{and} \quad y = t^2 \quad \text{for } t \geq 0$$

The point $P$ lies on the $y$-axis. The curve meets the $y$-axis when $t = 0$ and at point $P$.




a) Find the value of $t$ at $P$.
b) Find the area of the shaded region, which is bounded by the curve, the positive $x$-axis, the lines $x = 0$ and $x = 4$.

**Question 7.** A curve is defined by parametric equations

$$x = \frac{t}{t+4} \qquad \text{and} \qquad y = \frac{8}{(t+4)^2}$$

where $0 \leq t \leq 5$.

a) Find the coordinates of the curve when $t = 0$ and when $t = 5$.

b) Show that the area enclosed by the curve, the axes and the line $x = \frac{5}{9}$ is given by

$$\int_0^5 \frac{32}{(t+4)^4}\, dt$$

c) Hence, find the exact area.

**Question 8.** The parametric equations of a curve are

$$x = \tan\theta \quad \text{and} \quad y = \cos^3\theta$$

where $0 \leq \theta < \frac{\pi}{2}$.

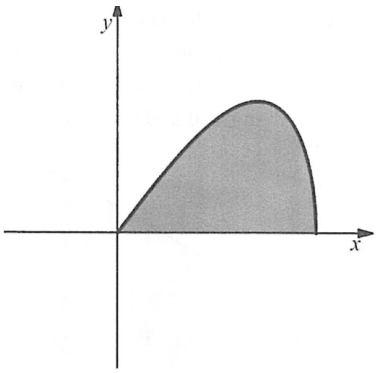
a) Find the coordinates of the curve when $x = 0$ and when $x = \sqrt{3}$.

b) Show that the area enclosed by the curve, the $x$-axis and the line $x = \sqrt{3}$ is $\frac{\sqrt{3}}{2}$.

**Question 9.** The diagram shows part of the curve defined by parametric equations

$$x = 3\sin t \quad \text{and} \quad y = 2\sin 2t$$

for $0 \leq t \leq \frac{\pi}{2}$.




The shaded region is bounded by the curve and the $x$-axis.

a) Show that the shaded area is given by

$$\int_0^{\frac{\pi}{2}} 12\sin t\cos^2 t\, dt$$

b) Hence show that the shaded area has an exact area of 4.

**Question 10.** A curve is defined by the parametric equations

$$x = t^3 \quad \text{and} \quad y = k\sqrt{t}$$

where $0 \leq t \leq 1$ and $k$ is a positive constant. It is known that the area between the curve, the $x$-axis and the lines $x = 0$ and $x = 1$ is 3.

a) Show that $k\int_0^1 3t^{\frac{5}{2}}\, dt = 3$.

b) Find the value of $k$.

**Question 11.** A curve is defined by parametric equations

$$x = t + \sin t \qquad \text{and} \qquad y = \tfrac{1}{2}\cos^2 t$$

where $0 \leq t \leq \pi$.

a) Find the coordinates of the curve when $t = 0$ and when $t = \pi$.

b) Show that the area enclosed by the curve, the axes and the line $x = \pi$ is given by

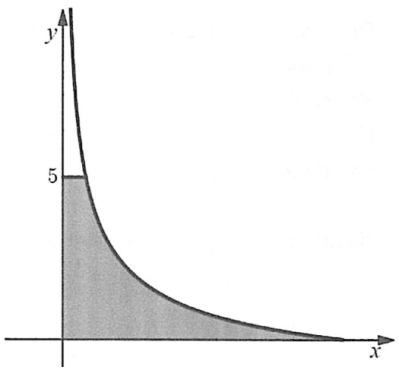
$$\frac{1}{2}\int_0^{\pi} (1 - \sin^2 t + \cos t - \cos t \sin^2 t)\, dt$$

c) Hence, find the exact area.

**Question 12.** The diagram shows part of the curve defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = \frac{6}{t} - 2$$

for $0 \leq t \leq 3$.




The shaded region is bounded by the curve, the coordinate axes and the line $y = 5$.

a) By considering $\int_0^5 x\, dy$, show that the shaded area is given by

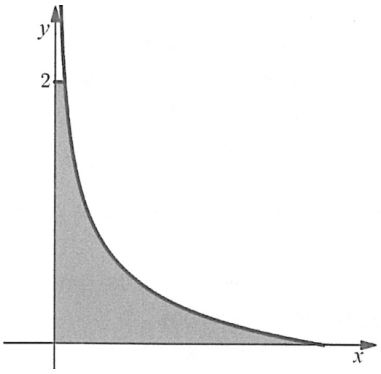
$$\int_{\frac{6}{7}}^{3} 6\, dt$$

b) Hence show that the shaded area has an exact area of $\frac{90}{7}$.

**Question 13.** The diagram shows part of the curve defined by parametric equations

$$x = 5t^3 \quad \text{and} \quad y = \frac{1}{t} - 1$$

for $0 \leq t \leq 1$.




The shaded region is bounded by the curve, the coordinate axes and the line $y = 2$.

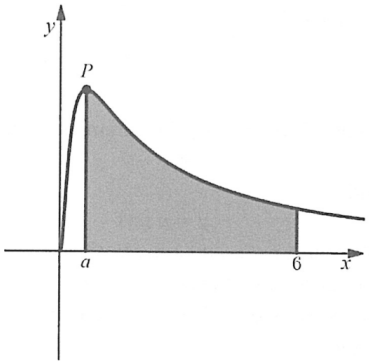
a) Show that the shaded area is given by $-\int_{\frac{1}{3}}^{1} 5t\,dt$.

b) Hence show that the shaded area has an exact area of $\frac{20}{9}$.

**Question 14. Prerequisite knowledge for Q14–Q22: Double Angle Formulae, Integration by Parts, Product Rule, Partial Fractions, Integration by Substitution** The diagram shows the curve given by

$$x = \frac{6}{t^2} \quad \text{and} \quad y = t^3 e^{-t}$$

where $t > 0$. The maximum of the curve is at the point $P$ with $x = a$ where $a$ is a constant to be found.




a) Find the coordinates of $P$.

b) Hence, show that the area enclosed by the curve, the $x$-axis and the line $x = a$ and $x = 6$ is

$$\frac{12}{e}\left(1 - \frac{1}{e^2}\right)$$

**Question 15.** Find the area under the curve defined parametrically by

$$y = \cos\theta, \quad x = \sin\theta$$

between $\theta = 0$ and $\theta = \frac{\pi}{2}$.

**Question 16.** Given

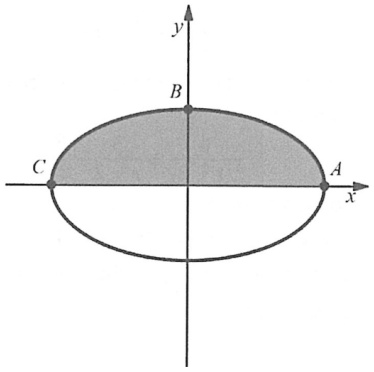
$$x = 2\sin t \quad \text{and} \quad y = 4\cos t$$

find $\int_0^2 y\,dx$ giving your answer in exact form.

**Question 17.** The diagram shows the curve defined by parametric equations

$$x = 9\cos\theta \quad \text{and} \quad y = 5\sin\theta$$

for $0 \leq \theta < 2\pi$.




The curve meets the $x$-axis at points $A$ and $C$. The curve meets the positive $y$-axis at the point $B$.

a) Find the coordinates of $A$, $B$ and $C$ and hence the value of $\theta$ at each of these points.

b) Show that the shaded region enclosed by the ellipse is given by

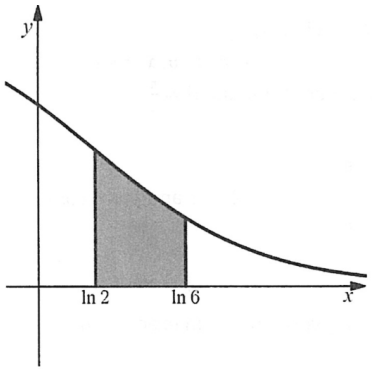
$$\int_0^{\pi} -45\sin^2\theta\, d\theta$$

c) Hence show that the shaded region has area $\frac{45\pi}{2}$.

**Question 18.** The diagram shows the curve defined by the parametric equations

$$x = \ln(t-2) \qquad \text{and} \qquad y = \frac{6}{t}$$

for $t \geq 2$.




The shaded region is closed by the curve, the $x$-axis and the lines $x = \ln 2$ and $x = \ln 6$.

a) Show that the shaded area may be given by

$$\int_{t_1}^{t_2} \frac{6}{t(t-2)}\, dt$$

where $t_1$ and $t_2$ are constants to be found.

b) Show that

$$\frac{6}{t(t-2)} \equiv \frac{A}{t-2} + \frac{B}{t}$$

for some constants $A$ and $B$.

c) Hence find the exact value of the shaded area in the form $\ln(\frac{a}{b})$ where $a$ and $b$ are positive integers.

**Question 19.** The parametric equations of a curve are

$$x = \frac{2t^2}{2+t} \qquad \text{and} \qquad y = \frac{1}{2(2+t)}$$

where $t \geq 0$.

a) Find the coordinates of the curve when $x = 0$ and when $x = \frac{18}{5}$.

b) Show that the area enclosed by the curve, the $x$-axis and the line $x = 9$ is given by

$$\int_0^3 \frac{t(4+t)}{(2+t)^3}\,dt$$

c) Hence, by using the substitution $u = 2 + t$ find the exact area.

**Question 20.** A curve is defined by parametric equations

$$x = t^2 \quad \text{and} \quad y = 1 - e^{-t}$$

for $t \geq 0$.

a) Find the $y$-value of the curve when $x = 0$ and when $x = 4$.

b) Sketch the curve for $0 \leq x \leq 4$.

c) Show that the area enclosed by the curve, the $x$-axis and the line $x = 4$ is given by

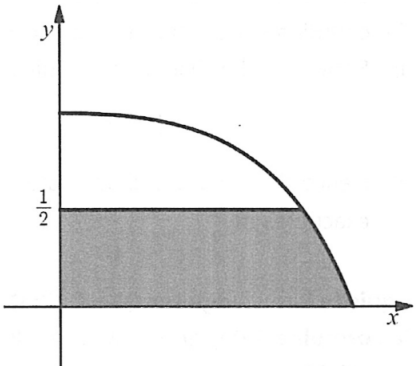
$$\int_0^2 (2t - 2te^{-t})\,dt$$

d) Hence, show the exact area is $2 + \frac{6}{e^2}$.

**Question 21.** The diagram shows part of the curve defined by parametric equations

$$x = \frac{t}{t+1} \quad \text{and} \quad y = 1 - t^2$$

for $0 \leq t \leq 1$.




The shaded region is bounded by the curve, the coordinate axes and the line $y = \frac{1}{2}$.

a) Show that the shaded area is given by

$$\int_{\frac{\sqrt{2}}{2}}^{1} \frac{2t^2}{t+1}\, dt$$

b) Hence, by using the substitution $u = t + 1$ show that the shaded area has an exact area of

$$-\frac{3}{2} + \sqrt{2} + 2\ln(4 - 2\sqrt{2})$$

**Question 22.** A curve is defined by the parametric equations

$$x = t + t^2 \qquad \text{and} \qquad y = k \sin t$$

where $0 \leq t \leq \pi$ and $k$ is a positive constant. It is known that the area between the curve, the $x$-axis and the lines $x = 0$ and $x = 12$ is 10. Show that

$$k = \frac{10}{1 + 2\sin 3 - 7\cos 3}$$